

DESIGN THINKING FOR SOFTWARE ENGINEERING

(LAB – 3 REPORT)

TOPIC : E-LEARNING PLATFORMS FAIL TO KEEP LEARNERS ENGAGED

Aim of the Experiment

The aim of this experiment is to apply the Design Sprint methodology (Empathize, Define, Ideate, Prototype, and Test) to design a user-centered solution that improves engagement in online learning platforms. The experiment focuses on understanding real learner problems across different age groups and validating solutions through user feedback rather than assumptions.

Problem Context

Online learning platforms are widely used by students of different age groups, but many learners struggle to stay engaged. Long lectures, lack of interaction, limited feedback, and monotonous content lead to boredom, distraction, and incomplete courses. These issues affect school students, undergraduate learners, and job-seeking students differently, making engagement a critical design challenge.

STEP 1: Empathize (Revisit & Deepen User Understanding)

Objective

To understand learners' emotions, frustrations, behavior, and needs while using online learning platforms.

Activities Performed

- Revisited user personas created in Lab 2 (Aishwarya, Yuktha, Manoj)
- Observed how learners interact with online classes and recorded videos
- Noted emotional reactions such as boredom, confusion, and motivation loss

Key Observations



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- Learners lose focus after 15–20 minutes
- Younger learners need visual and interactive content
- Older learners prefer practical, outcome-oriented learning
- Lack of feedback reduces confidence

Refined User Insights

- Engagement depends more on interaction than content length
 - One design approach does not fit all age groups
 - Learners want learning to feel active, not passive
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STEP 2: Define (Problem Statement & Design Challenge)

Problem Statement

Online learners need an engaging and interactive way to learn because long, passive, and non-personalized content reduces attention, motivation, and course completion.

How Might We (HMW) Question

How might we design an online learning experience that keeps learners of different age groups engaged, motivated, and confident throughout the course?

STEP 3: Ideate (Idea Generation)

Brainstormed Ideas

- Short 5–10 minute video lessons
- Interactive quizzes after every topic
- Gamification with points, badges, and rewards
- Live doubt-clearing chat sessions
- Practice-based mini tasks and challenges
- Personalized learning paths based on age and goals

Idea Selection

Selected Idea:

An **Interactive Online Learning Platform** with short lessons, quizzes, gamification, and age-based personalization.

Justification



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- High engagement value
 - Feasible with existing technology
 - Addresses pain points of all three personas
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STEP 4: Prototype

Objective

To create a low-fidelity prototype to represent the selected solution.

Prototype Description

The prototype includes:

- Dashboard with progress tracker
- Short video lesson screen (5–10 minutes)
- Quiz after each lesson
- Rewards (points, badges)
- Practice tasks for skill application

Design Rationale

- Short lessons reduce boredom
- Quizzes improve attention and retention
- Gamification increases motivation
- Progress tracking builds confidence

Prototype Type

Low-fidelity wireframes / paper sketches

STEP 5: Test (User Feedback & Validation)

Testing Method

- Prototype concept explained to three learners (one for each persona)
- Learners were asked to simulate course usage
- Feedback and reactions were noted

User Feedback Summary

User	What Worked	What Confused	Suggestions
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User	What Worked	What Confused	Suggestions
Aishwarya (19)	Short videos, quizzes	Navigation	Add reminders
Yuktha (12)	Games, visuals	Text-heavy screens	More animations
Manoj (23)	Practice tasks	Limited advanced content	Add real projects

Observations

- Learners stayed attentive longer
- Motivation increased due to rewards
- Confidence improved with instant feedback

STEP 6: Iterate & Reflect

Improvements Suggested

- Add adaptive difficulty levels
- Include live mentor interaction
- Improve personalization based on learner goals

Reflection

This experiment showed that learners are not disengaged because they dislike learning, but because current platforms are passive and generic. User feedback helped refine the solution and highlighted the importance of interaction, personalization, and motivation.

Conclusion

The Design Sprint experiment successfully applied empathy-driven design to improve engagement in online learning platforms. By understanding learners' real needs and validating ideas through feedback, a practical and user-centered solution was developed. This experiment demonstrates that interactive and learner-focused design is essential for effective e-learning systems.

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